

Vardhan Thatikonda

Data Scientist

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Profile

Accomplished Data Scientist with 3+ years of experience excelling in the utilization of data to inform strategic business decisions. Proficient in employing advanced statistical analysis and cutting-edge machine learning techniques to derive actionable insights from intricate datasets. Adept at designing and implementing predictive models, conducting comprehensive data exploration, and creating scalable data pipelines.

Technical Skills

- **Methodologies and IDEs:** SDLC, Agile, Waterfall, Visual Studio Code, PyCharm, Colab
- **Language and Databases:** Python, Go, SQL, R, C/C++, MATLAB, Scala, MySQL, Oracle, Firebase, MongoDB, Redshift,
- **Library/ Packages:** PyTorch, TensorFlow, Keras, NumPy, SciPy, Pandas, Regex, SciKit-Learn, XGBoost, OpenCV, NLTK, SpaCy, Matplotlib, Ggplot, Seaborn, ResNet 50
- **Tools:** Power BI, Tableau, Microsoft Excel, Hive, Alteryx, Spark, Airflow, Snowflake, Hadoop, MLflow, Docker, Kubernetes, Google Analytics, Git, GitHub, Jira, and Jenkins, ETL
- **ML Algorithms:** Regression, Supervised Learning, Unsupervised Learning, Random Forest, Linear Regression, Decision Tree, Deep Learning, Clustering, Classification, Time-Series, Tensorflow, Keras, NLP, GANs, Open AI, LLM, RNN, CNN
- **Other Skills:** Data Cleaning, Data Wrangling, Data Warehousing, Data Visualization, Critical Thinking, Communication Skills, Presentation Skills, Problem Solving, Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform, A/B Testing

Professional Experience

Data Scientist, PTC

08/2024 – Present | USA

- Developed and deployed a personalized recommendation system utilizing collaborative filtering, content-based filtering, autoencoder models, and LLMs, resulting in a 15% increase in sales, improved customer satisfaction, and enhanced marketing effectiveness.
- Implemented K-means clustering for customer segmentation, enhancing recommendation precision by tailoring product offerings to specific customer groups based on behavior patterns and preferences.
- Conducted comprehensive data preparation using Python and SQL to clean, transform, and create a robust database structure, ensuring high-quality data for accurate model training and analysis.
- Engineered features and optimized machine learning models, including an autoencoder-based recommendation model that achieved 92% accuracy in predicting user preferences, utilizing cross-validation and hyperparameter tuning techniques with Python libraries (Pandas, NumPy, Scikit-learn).
- Deployed models over AWS Cloud, utilizing AWS SageMaker for training, Lambda for serverless deployment, Amazon Redshift for data warehousing, and CloudWatch and Prometheus for real-time monitoring and performance tracking.

Data Scientist Intern, Kofax

08/2023 – 05/2024 | USA

- Implemented a predictive model using Python, Pandas, NumPy, and Scikit-learn, resulting in a substantial decrease in stockouts for essential medical supplies, thereby enhancing patient care and operational reliability.
- Collected, cleaned, and preprocessed extensive datasets from Electronic Health Records (EHRs) and hospital management systems, utilizing advanced data manipulation with Python and Excel to ensure high-quality data inputs for modeling.
- Engineered a high-accuracy hybrid forecasting model by integrating time-series analysis, ARIMA with machine learning algorithms (Random Forest, XGBoost, TensorFlow/Keras), which was rigorously validated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) metrics, achieving a high degree of predictive accuracy.
- Evaluated model performance through machine learning metrics and deployed the solution on AWS (EC2 and S3), ensuring scalability and seamless integration with the hospital's existing supply chain system.
- Integrated real-time inventory recommendations and automated purchase order generation into the hospital's supply chain management system, significantly improving operational efficiency and ensuring timely availability of supplies.

Data Scientist, Sigma Info Tech

02/2020 – 07/2022 | India

- Developed a sophisticated predictive model for property sale prices using a dataset with over 10,000 features, leveraging machine learning algorithms (Linear Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM) and deep learning frameworks (MLPs, RNNs) with TensorFlow and PyTorch.
- Conducted extensive data preprocessing on real estate text data from platforms like Zillow and Realtor.com, addressing 5% missing values through imputation, normalization, and standardization, while enhancing feature sets using advanced NLP techniques such as tokenization, stemming, lemmatization, and sentiment analysis.
- Designed and implemented ETL processes to efficiently extract data from multiple sources, transform it for analysis, and load it into a structured format, ensuring data integrity and accessibility for modeling.
- Conducted A/B testing to compare model performance across different algorithms and configurations, enabling data-driven decisions on the most effective approach for predicting property sale prices.
- Created over 20 interactive and static visualizations using Matplotlib, Seaborn, and Plotly to illustrate complex data patterns and model results, facilitating data-driven decision-making.

Education

Masters of Science, University of Texas at Arlington

08/2022 - 05/2024 | Arlington, USA

Data Science

Bachelor of Technology, Saveetha Engineering College

08/2017 - 04/2021 | Chennai, India

Electronics and Communication Engineering